Lecture 25

STAT 109: Introductory Biostatistics

Lecture 25: One numerical variable across two groups: spread, boxplots, and two-sample *t*

Learning objectives

- Compute **range**, **quartiles**, **IQR**, and **standard deviation** by hand and compare two small datasets.
- Build and interpret **boxplots** as a picture of the **five-number summary**.
- State the **Welch two-sample *t*-test** (unequal variances), its t-statistic, **degrees of freedom**, hypotheses, and conditions; run it in R with `t.test(y ~ group)`.
- Use `group_by()` with `summarize()` for descriptives and **ggplot2** (`fill` or `color`) to compare groups with histograms, dot plots, or boxplots.

Setting: one numerical variable, two categories

We often compare a **numerical** outcome (e.g. recovery time, blood pressure) between two groups defined by a **binary categorical** variable (treatment vs. control, exposed vs. not exposed). Descriptively we compare center and spread; inferentially we may compare population means with a **two-sample *t*-test**.

Two small datasets (hand practice, $n = 8$ each)

We use eight values per group (within the usual 7–9 range for classroom hand calculations) so that sample SD is a clean calculation and quartiles split evenly into halves of four.

Dataset N (“narrow”): $7,\ 8,\ 9,\ 10,\ 10,\ 11,\ 12,\ 13$

Dataset W (“wide”): $4,\ 6,\ 8,\ 10,\ 10,\ 12,\ 14,\ 16$

Both have mean $\bar{x} = 10$ (check: sums are 80 each). The spread differs: N is clustered near 10; W stretches toward 4 and 16.

Range

- N: $\max - \min = 13 - 7 = 6$.
- W: $16 - 4 = 12$.

Standard deviation (sample)

First, with **no sigma notation**: for n observations $x_1, x_2, \dots, x_n$ with mean $\bar{x}$, the **sample standard deviation** is

$$s = \sqrt{\frac{1}{n-1}\left[(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2\right]}.$$

You average the **squared deviations** from the mean (add them all, then divide by $n - 1$), then take the **square root** so the result is on the same scale as the data.

The same quantity in **sigma (summation) notation** is

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Here $\sum_{i=1}^{n}(x_i - \bar{x})^2$ means "add $(x_i - \bar{x})^2$ for $i = 1, 2, \dots, n$," i.e. the numerator inside the square root in the expanded form above.

By hand for dataset N ($n = 8$, $\bar{x} = 10$). Put the data in column 1, then add columns for the deviation from the mean and its square. Total column 3 to get the **sum of squared deviations** (often called **SS**).

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
7	-3	9
8	-2	4
9	-1	1
10	0	0
10	0	0
11	1	1
12	2	4
13	3	9

Sum of column 3: $9 + 4 + 1 + 0 + 0 + 1 + 4 + 9 = 28$.

The **sample variance** is the average of the squared deviations using divisor $n - 1$:

$$s^2 = \frac{28}{8-1} = \frac{28}{7} = 4.$$

(Aside: s^2 is the **variance**; it has nicer algebra but is in squared units.) The **sample SD** is the square root:

$$s = \sqrt{4} = 2.$$

Dataset W ($\bar{x} = 10$). Repeat the same steps: column 1 is x, column 2 is $x - \bar{x}$, column 3 is $(x - \bar{x})^2$, then sum column 3 and finish with variance and SD. Use the space below in the PDF.

Answer (dataset W).

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-6	36
6	-4	16
8	-2	4
10	0	0
10	0	0
12	2	4
14	4	16
16	6	36

Sum of column 3: $36 + 16 + 4 + 0 + 0 + 4 + 16 + 36 = 112$.

$$s^2 = \frac{112}{7} = 16, \qquad s = \sqrt{16} = 4.$$

Same mean, different SD, W is twice as variable as N in this sense.

Interpreting SD

A single **s**reported SD is hard to interpret on its own: it is measured in the same units as the data, but without context you do not know whether "2" or "4" is large or small for that variable. SD is most useful when you compare the spread of the same numerical outcome across two or more groups (or to another study using the same scale), as with N vs. W above.

If you are willing to assume the population is approximately normal, a lone SD becomes more informative because of the empirical rule: roughly 68% of values fall within one SD of the mean, 95% within two SDs, and 99.7% within three SDs. Outside that setting, treat a standalone SD as a building block for comparisons or for modeling, not as a full story by itself.

Optional: mean and median absolute deviation from the median

Two related summaries use **absolute** (not squared) distances from the **median** $\tilde{x}$. They are easy to interpret for **one** group or for **comparing** groups.

Mean absolute deviation from the median (average distance to the median):

$$\frac{1}{n}\sum_{i=1}^{n} |x_i - \tilde{x}|.$$

Median absolute deviation (often written **MAD**): take the median of the n numbers $|x_1 - \tilde{x}|, \ldots, |x_n - \tilde{x}|$. Half of those absolute deviations are at or below MAD and half are at or above, so MAD is a typical magnitude of distance from the median (robust to outliers in the same spirit as the median itself).

For **dataset N**, $\tilde{x} = 10$. The absolute deviations $|x_i - 10|$ are $3, 2, 1, 0, 0, 1, 2, 3$.

- Mean absolute deviation from the median: $(3 + 2 + 1 + 0 + 0 + 1 + 2 + 3)/8 = 12/8 = 1.5$.
- Median absolute deviation: sorted deviations $0, 0, 1, 1, 2, 2, 3, 3$, so MAD $= (1 + 2)/2 = 1.5$.

For **dataset W**, $\tilde{x} = 10$. The values $|x_i - 10|$ are $6, 4, 2, 0, 0, 2, 4, 6$.

- Mean absolute deviation from the median: $(6 + 4 + 2 + 0 + 0 + 2 + 4 + 6)/8 = 24/8 = 3$.
- Median absolute deviation: sorted $0, 0, 2, 2, 4, 4, 6, 6$, so MAD $= (2 + 4)/2 = 3$.

Interpretation. The **mean** absolute deviation from the median answers: on **average**, how far are observations from the median? Here, about **1.5** units (N) vs **3** units (W), matching the idea that W is more spread out. The **MAD** gives a **typical** absolute distance from the median (half of the absolute deviations fall at or below it, half at or above); for N and W it matches the mean absolute deviation here because the deviation lists are symmetric. Either measure is useful alone or side by side with SD when you compare groups.

Quartiles and IQR by Hand

Sort the data. For eight values, the median is the average of the 4th and 5th values. Split into lower four and upper four; Q_1 is the median of the lower half, Q_3 the median of the upper half.

N: sorted $7, 8, 9, 10, 10, 11, 12, 13$. Median $= (10 + 10)/2 = 10$. Lower half $7, 8, 9, 10 \Rightarrow Q_1 = (8 + 9)/2 = 8.5$. Upper half $10, 11, 12, 13 \Rightarrow Q_3 = (11 + 12)/2 = 11.5$. IQR $= 11.5 - 8.5 = 3$.

W: sorted $4, 6, 8, 10, 10, 12, 14, 16$. Median $= 10$. Lower half $4, 6, 8, 10 \Rightarrow Q_1 = 7$. Upper half $10, 12, 14, 16 \Rightarrow Q_3 = 13$. IQR $= 6$.

Second example (odd n, outliers). Take **nine** values so the **median** is a **single** order statistic (the middle of the sorted list). Use the same rule as above: sort the data, take the **median**, then form the **lower** and **upper** halves **excluding** the median (here each half has **four** values), and take the median of each half as Q_1 and Q_3. For four numbers, the median is the **average of the two middle** values in that half.

Dataset A (symmetric): 5, 6, 7, 8, 9, 10, 11, 12, 13. Sorted: $5, 6, 7, 8, 9, 10, 11, 12, 13$.

- $n = 9$ is **odd**, so the **median** is the **5th** value: $\tilde{x} = 9$.
- **Lower half** (four values below the median): $5, 6, 7, 8 \rightarrow Q_1 = (6 + 7)/2 = 6.5$.
- **Upper half** (four values above the median): $10, 11, 12, 13 \rightarrow Q_3 = (11 + 12)/2 = 11.5$.
- **IQR** $= 11.5 - 6.5 = 5$.

Dataset B (two large values in the upper tail): 5, 6, 7, 8, 9, 10, 11, 90, 100. Sorted: $5, 6, 7, 8, 9, 10, 11, 90, 100$.

- The **median** is still the **5th** value: $\tilde{x} = 9$.
- **Lower half:** $5, 6, 7, 8 \rightarrow Q_1 = 6.5$ (unchanged from A).
- **Upper half:** $10, 11, 90, 100 \rightarrow Q_3 = (11 + 90)/2 = 50.5$ (the two middle values in the upper four are 11 and 90).
- **IQR** $= 50.5 - 6.5 = 44$.

So Q_3 and **IQR** jump when the **upper four** values include extremes that move the **middle pair** used for Q_3. With $n = 9$, a **single** outlier at the very top often **does not** change Q_3 (the upper half still has four values, and the middle two can stay small), which is why this illustration uses **two** large upper values. By contrast, **dataset N** and **W** above use **even** $n = 8$, where the overall median is the **average** of the two middle order statistics and each half has **four** observations. Same **Tukey-style** splitting idea.

Five-number summary: the values of min, Q_1, median, Q_3, max break the dataset into 4 roughly equally sized pieces.

Note: Software may use slightly different quartile interpolation; match your course's convention and `?quantile` in R if you need exact agreement.

Boxplots and the five-number summary

A **boxplot** shows the **five-number summary**: the box runs from Q_1 to Q_3 (spanning the middle 50%), the median is drawn inside, and whiskers extend to sensible extremes (often the furthest points within $1.5 \times$ IQR of the quartiles, with more extreme points as outliers). Rules differ by implementation, check `?geom_boxplot`.

For N and W, sketch two boxes side by side: same median (10), but W has a wider box and whiskers, matching larger IQR and range.

Check in R

Put both small datasets in **one** data frame: a **numerical** outcome `y` and a **binary categorical** `group` (`Narrow` vs `Wide`). Then use `group_by()` and `summarize()` for descriptives, and **ggplot2** for plots.

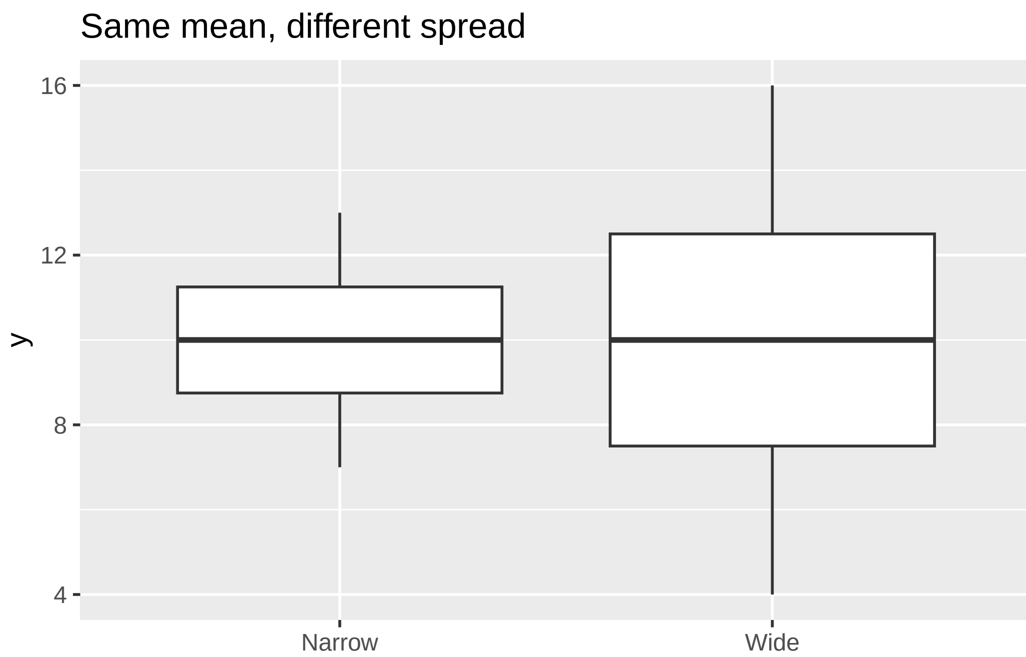
```r
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

compare <- tibble(
  group = factor(
    rep(c("Narrow", "Wide"), each = 8),
    levels = c("Narrow", "Wide")
  ),
  y = c(
    7, 8, 9, 10, 10, 11, 12, 13,
    4, 6, 8, 10, 10, 12, 14, 16
  )
)
```

```r
compare |>
  group_by(group) |>
  summarize(
    n = n(),
    mean_y = mean(y),
    sd_y = sd(y),
    median_y = median(y),
    min_y = min(y),
    max_y = max(y),
    range_y = max(y) - min(y),
    q1_R = quantile(y, 0.25, names = FALSE),
    q3_R = quantile(y, 0.75, names = FALSE),
    iqr = IQR(y),
    mad_from_median = median(abs(y - median(y))),
    mean_abs_dev_from_median = mean(abs(y - median(y)))
  )
```

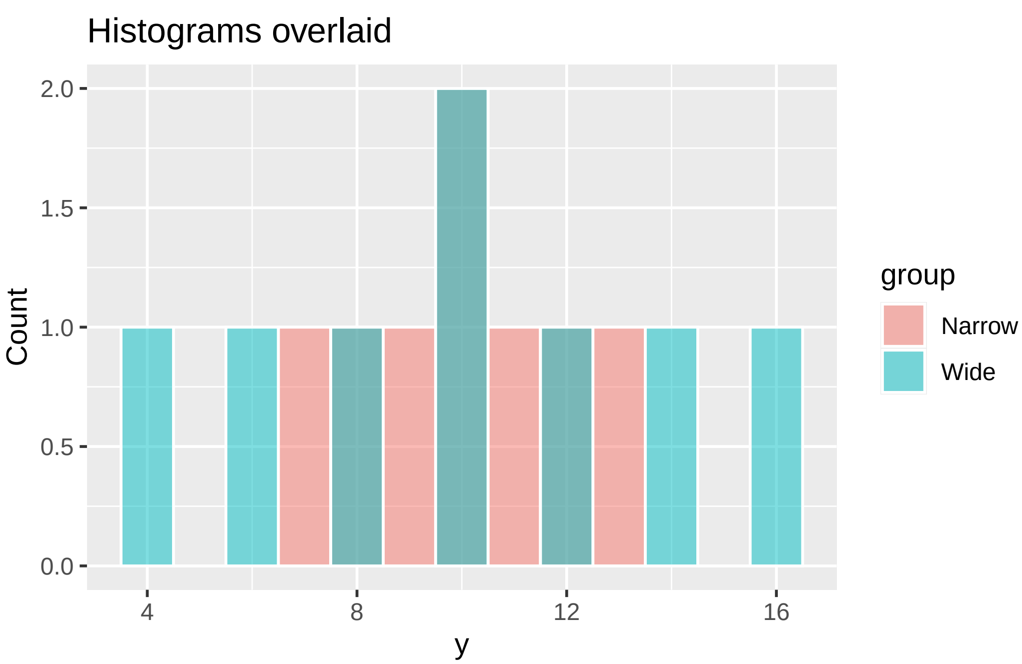
```
# A tibble: 2 x 13
  group      n mean_y  sd_y median_y min_y max_y range_y  q1_R  q3_R   iqr
  <fct>  <int>  <dbl> <dbl>    <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl>
1 Narrow     8     10     2       10     7    13       6  8.75  11.2   2.5
2 Wide       8     10     4       10     4    16      12  7.5   12.5   5
# i 2 more variables: mad_from_median <dbl>, mean_abs_dev_from_median <dbl>
```

`quantile()` and `IQR()` use **R**'s default definition (see `?quantile`), which may differ slightly from the **Tukey-style** hinges in the hand work above. **MAD** here is the **median** absolute deviation from the **median**; **mean_abs_dev_from_median** is the **mean** of those absolute deviations.

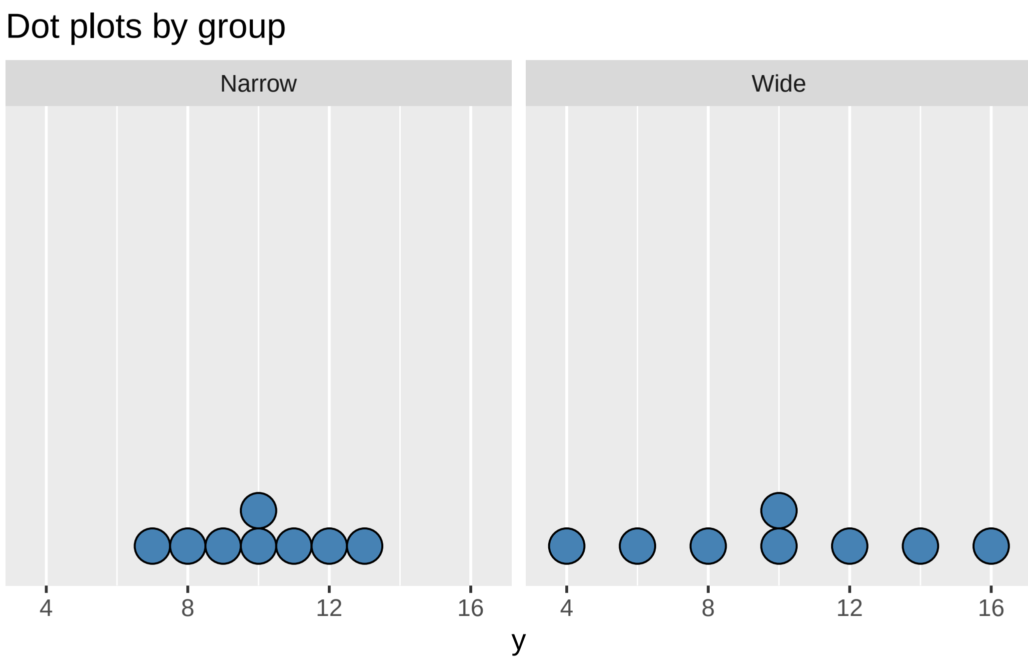
```r
ggplot(compare, aes(x = group, y = y)) +
  geom_boxplot() +
  labs(title = "Same mean, different spread", x = NULL, y = "y")
```




```r
ggplot(compare, aes(x = y, fill = group)) +
  geom_histogram(binwidth = 1, position = "identity", alpha = 0.5, color = "white") +
  labs(title = "Histograms overlaid", x = "y", y = "Count")
```




```r
ggplot(compare, aes(x = y)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "steelblue") +
  facet_wrap(~ group) +
  labs(title = "Dot plots by group", x = "y") +
  scale_y_continuous(NULL, breaks = NULL)
```



Two-sample t-test (independent groups, unequal variance)

Parameters. μ_1 and μ_2 are the population means of the numerical outcome in group 1 and group 2.

Hypotheses (compare means).

- H_0: $\mu_1 = \mu_2$ (often written $\mu_1 - \mu_2 = 0$).
- H_a: two-sided $\mu_1 \neq \mu_2$, or one-sided greater / less for $\mu_1 - \mu_2$.

The symbols μ_1 and μ_2 are **population** means: unknown quantities describing the two groups in the world or in a model. The hypotheses are **not** about the **sample** means $\bar{x}_1$ and $\bar{x}_2$ from the dataset (those are computed directly from the data). The test uses the sample to **draw conclusions about** μ_1 and μ_2.

Setting. Independent random samples (or a randomized experiment with two arms). Welch's procedure does not assume equal population SDs (it allows $\sigma_1 \neq \sigma_2$).

Test statistic (Welch). With sample means $\bar{x}_1$, $\bar{x}_2$ and sample SDs s_1, s_2 from sample sizes n_1, n_2,

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

Under H_0 and when conditions are reasonable, t is referred to a **Student t distribution** with degrees of freedom estimated by the Welch–Satterthwaite formula (not necessarily an integer). R reports this df; we do not compute it by hand.

Conditions (practical).

1. Independent observations between and within groups.
2. Sample size / shape: mild skew is often acceptable for moderate n; small samples and strong skew or outliers need caution.

Default in R: `t.test(y ~ group)` uses Welch (`var.equal = FALSE`).

Example: different means (same $n = 8$)

Group A: 7, 8, 9, 10, 10, 11, 12, 13 (sample mean $\bar{x}_A = 10$).

Group B: 9, 10, 11, 12, 12, 13, 14, 15 (sample mean $\bar{x}_B = 12$).

Those **sample** means are descriptive; the **test** still asks about μ_A and μ_B.

```
ex <- tibble(
  group = factor(rep(c("A", "B"), each = 8)),
  y = c(7, 8, 9, 10, 10, 11, 12, 13, 9, 10, 11, 12, 12, 13, 14, 15)
)

t.test(y ~ group, data = ex, var.equal = FALSE)
```

```

	Welch Two Sample t-test

data:  y by group
t = -2, df = 14, p-value = 0.06529
alternative hypothesis: true difference in means between group A and group B is not equal to 0
95 percent confidence interval:
 -4.1447867  0.1447867
sample estimates:
mean in group A mean in group B 
             10              12 
```

R reports t, df, a **p-value** for the default **two-sided** alternative H_a: $\mu_A \neq \mu_B$, and a **confidence interval** for $\mu_A - \mu_B$ (difference in **population** means).

Decision rule (significance level 0.05**).** If the **p-value is less than** 0.05, reject H_0: $\mu_A = \mu_B$ in favor of H_a: $\mu_A \neq \mu_B$, i.e. conclude there is evidence that the **population** means of y in the two groups **differ**. That rule is set up so that, when H_0 is **actually true**, you reject H_0 at most **5%** of the time in the long run: that cap on the probability of a **Type I error** (deciding there is a difference in **population** means when there is none) is the **significance level** (here 0.05). If the p-value is **not** below 0.05, you do not reject H_0 at that level (you do not conclude "no difference," only "not enough evidence" at 5%).

`group_by()` descriptives and ggplot with `fill`

Check in R above already used `group_by(group) |> summarize(...)` on `compare` and mapped `fill` / `facet_wrap` in **ggplot2**. The same pattern applies to any tibble with one numeric variable and one grouping variable.

Syntax note. As in Lecture 24, `summarize()` and `summarise()` are equivalent, and `|>` matches `%>%` for piping; this lecture uses `|>` and `summarize()`.

For the **two-sample *t*-test** example, `ex` contains groups A and B. A compact summary:

```
ex |>
  group_by(group) |>
  summarize(
    n = n(),
    mean_y = mean(y),
    sd_y = sd(y),
    median_y = median(y)
  )
```

```
# A tibble: 2 x 5
  group     n mean_y  sd_y median_y
  <fct> <int>  <dbl> <dbl>    <dbl>
```

```
1 A          8      10     2        10
2 B          8      12     2        12
```

Reuse the **ggplot** calls from **Check in R** with `ex` in place of `compare` if you want the same plots for this example. Use `position = "dodge"` in bar-like geoms when categories would overlap; for histograms, `position = "identity"` with `alpha` or `facet_wrap(~ group)` often reads more clearly than a busy overlap.